Assignment 5: Data Visualization

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Fall 2025

OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Visualization

Directions

1. Rename this file `<FirstLast>_A05_DataVisualization.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Be sure your code is tidy; use line breaks to ensure your code fits in the knitted output.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.

Set up your session

1. Set up your session. Load the tidyverse, here & cowplot packages, and verify your home directory. Read in the NTL-LTER processed data files for nutrients and chemistry/physics for Peter and Paul Lakes (use the tidy `NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv` version in the Processed_KEY folder) and the processed data file for the Niwot Ridge litter dataset (use the `NEON_NIWO_Litter_mass_trap_Processed.csv` version, again from the Processed_KEY folder).

2. Make sure R is reading dates as date format; if not change the format to date.

```
#1
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.4     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(here)
```

```
## here() starts at /home/guest/EDE_Fall2025
```

```r
library(cowplot)
```

```
##
## Attaching package: 'cowplot'
##
## The following object is masked from 'package:lubridate':
##
##     stamp
```

```r
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
here()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
NTLLTER <-read.csv(here(
  './Data/Processed_KEY/NTL-LTER_Lake_Chemistry_Nutrients_PeterPaul_Processed.csv'),stringsAsFactors = T
NEON_NIWO <-read.csv(here(
  './Data/Processed_KEY/NEON_NIWO_Litter_mass_trap_Processed.csv'),
  stringsAsFactors = TRUE)

#2
NTLLTER$sampledate <- ymd(NTLLTER$sampledate)
NEON_NIWO$collectDate <- ymd(NEON_NIWO$collectDate)
class(NTLLTER$sampledate)
```

```
## [1] "Date"
```

```r
class(NEON_NIWO$collectDate)
```

```
## [1] "Date"
```

Define your theme

3. Build a theme and set it as your default theme. Customize the look of at least two of the following:

- Plot background
- Plot title
- Axis labels
- Axis ticks/gridlines
- Legend

```r
#3
devtools::install_github("katiejolly/nationalparkcolors")
```

```
## Skipping install of 'nationalparkcolors' from a github remote, the SHA1 (df8cd15d) has not changed si
##   Use `force = TRUE` to force installation
```

```r
library(nationalparkcolors)

names(park_palettes)
```

```
##  [1] "SmokyMountains" "RockyMountains" "Yellowstone"    "Arches"
##  [5] "ArcticGates"    "MtMckinley"     "GeneralGrant"   "Hawaii"
##  [9] "CraterLake"     "Saguaro"        "GrandTeton"     "BryceCanyon"
## [13] "MtRainier"      "Badlands"       "Redwoods"       "Everglades"
## [17] "Voyageurs"      "BlueRidgePkwy"  "Denali"         "GreatBasin"
## [21] "ChannelIslands" "Yosemite"       "Acadia"         "DeathValley"
## [25] "Zion"
```

```r
Yosemite_color <- park_palette("Yosemite")
#view(Yosemite_color)

Elizabeth_theme <- theme_grey(base_size = 10) +
  theme(
    plot.title = element_text( # Plot title
      size = 12,          #bigger title font
      face = "bold",      #Bold title
      hjust = 0.5,        #Centered title
      margin = margin(b = 12) #Add space below the title
    ),

    # Axis text
    axis.text = element_text(color = "black"),  #Black axis tick labels

    # Legend styling
    legend.position = "right",                  #Legend on the right
    legend.title = element_text(
      face = "bold"                            #Bold legend title
    ),
    legend.background = element_rect(
      color = "black",                          #black border
      fill = "white"                           #white background
    )
  )

theme_set(Elizabeth_theme)
```

Create graphs

For numbers 4-7, create ggplot graphs and adjust aesthetics to follow best practices for data visualization. Ensure your theme, color palettes, axes, and additional aesthetics are edited accordingly.

4. [NTL-LTER] Plot total phosphorus (`tp_ug`) by phosphate (`po4`), with separate aesthetics for Peter and Paul lakes. Add line(s) of best fit using the `lm` method. Adjust your axes to hide extreme values (hint: change the limits using `xlim()` and/or `ylim()`).

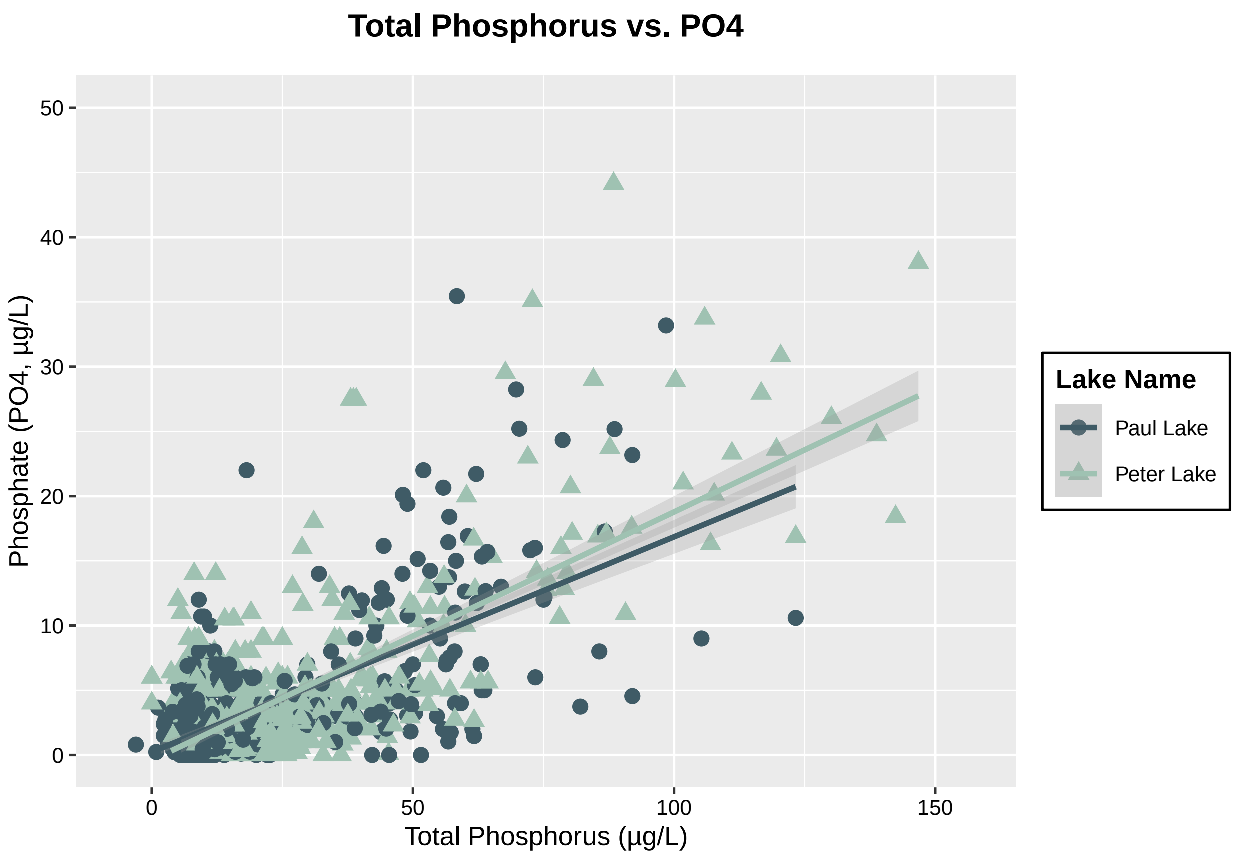
```
#4
TotalP.vs.PO4 <- ggplot(NTLLTER, aes(x = tp_ug, y = po4, shape = lakename, color = lakename)) +
  geom_point(size = 2.5) +
  geom_smooth(method = "lm", formula = y ~ x, se = TRUE,
              linewidth = 1, alpha = 0.25) +
  scale_color_manual(
    values = c("Peter Lake" = "#9FC2B2", "Paul Lake" = "#3F5B66")
  ) +
  labs(
    title = "Total Phosphorus vs. PO4",
    x = "Total Phosphorus (µg/L)",
    y = "Phosphate (PO4, µg/L)",
    color = "Lake Name",
    shape = "Lake Name"
  ) +
  ylim(0, 50) +
  Elizabeth_theme

print(TotalP.vs.PO4)
```

```
## Warning: Removed 21947 rows containing non-finite outside the scale range
## (`stat_smooth()`).

## Warning: Removed 21947 rows containing missing values or values outside the scale range
## (`geom_point()`).

## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_smooth()`).
```



5. [NTL-LTER] Make three separate boxplots of (a) temperature, (b) TP, and (c) TN, with month as the x axis and lake as a color aesthetic. Then, create a cowplot that combines the three graphs. Make sure that only one legend is present and that graph axes are aligned.

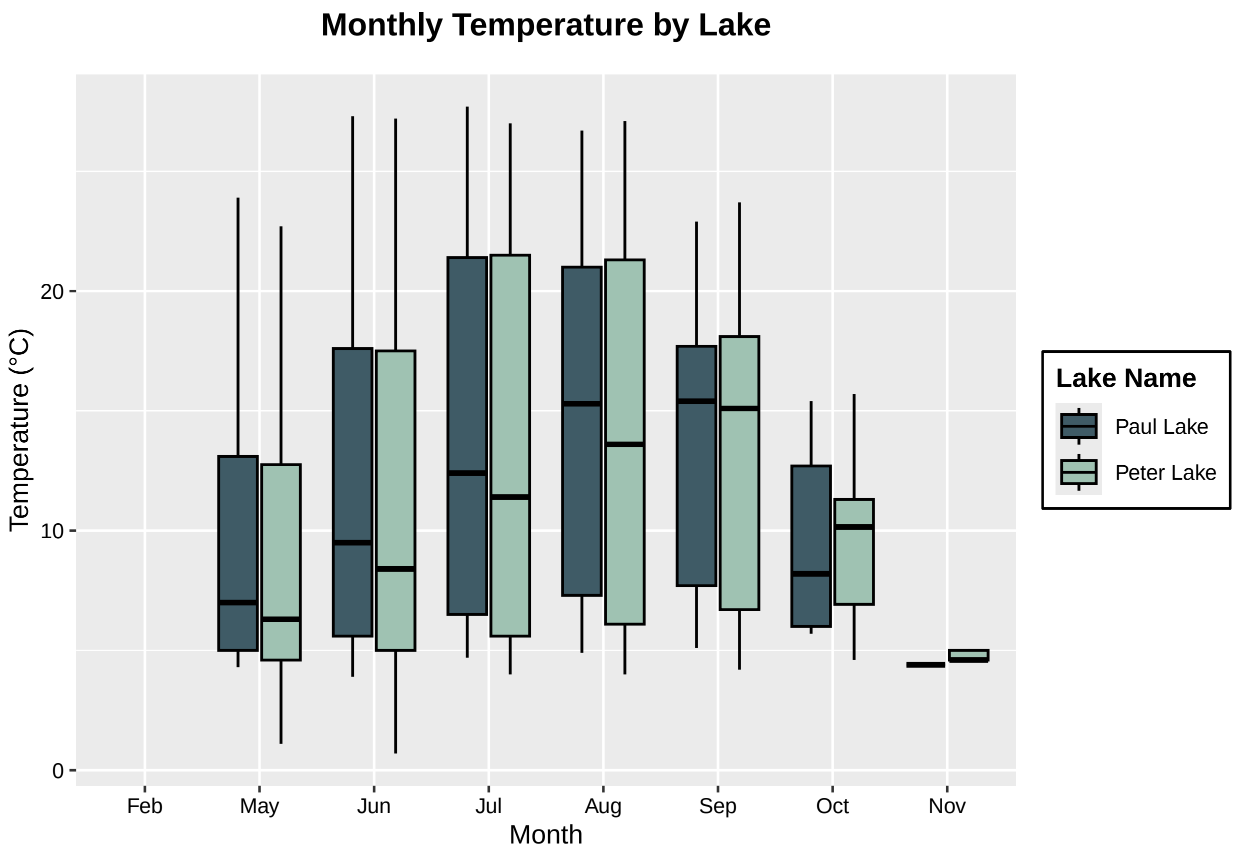
Tips: * Recall the discussion on factors in the lab section as it may be helpful here. * Setting an axis title in your theme to `element_blank()` removes the axis title (useful when multiple, aligned plots use the same axis values) * Setting a legend's position to "none" will remove the legend from a plot. * Individual plots can have different sizes when combined using `cowplot`.

```
#5
NTLLTER <- NTLLTER %>%
  mutate(
    month = factor(
      month(sampledate),
      levels = 1:12,
      labels = month.abb,
      ordered = TRUE))

Temp <- ggplot(NTLLTER, aes(x = month, y = temperature_C, fill = lakename)) +
  geom_boxplot(color = "black") +   # black outline for contrast
  labs(
    title = "Monthly Temperature by Lake",
    x = "Month",
    y = "Temperature (°C)",
    fill = "Lake Name"
```

```
  ) +
  scale_fill_manual(
    values = c("Peter Lake" = "#9FC2B2", "Paul Lake" = "#3F5B66")
  ) +
  Elizabeth_theme
print(Temp)
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```



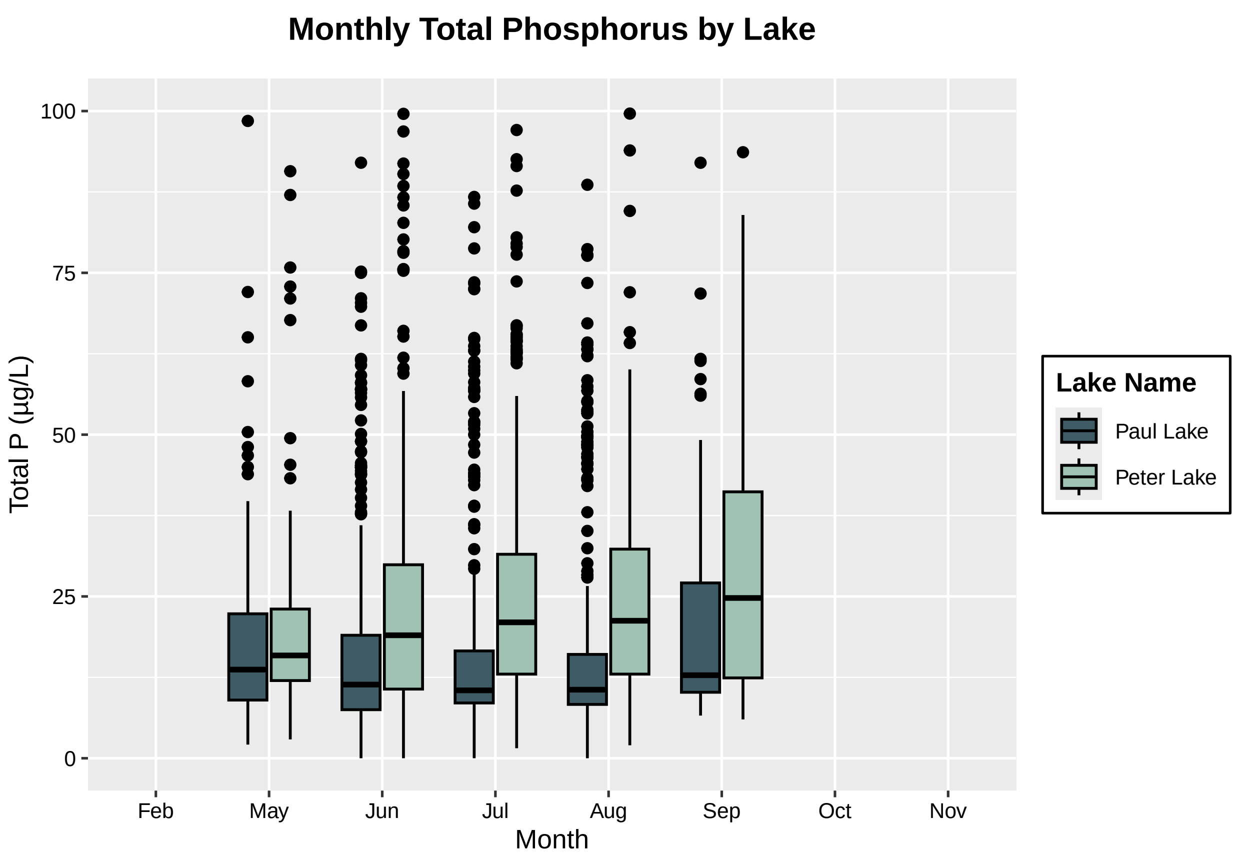

```
summary(NTLLTER$tp_ug)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##  -6.349   9.194  14.401  22.159  27.746 157.250   20729
```

```
TP <- ggplot(NTLLTER, aes(x = month, y = tp_ug, fill = lakename)) +
  geom_boxplot(color = "black") +
  labs(
    title = "Monthly Total Phosphorus by Lake",
    x = "Month",
    y = "Total P (µg/L)",
    fill = "Lake Name"
  ) +
```

```r
  scale_fill_manual(
    values = c("Peter Lake" = "#9FC2B2", "Paul Lake" = "#3F5B66")
  ) +
  ylim(0,100)+
  Elizabeth_theme
print(TP)
```

```
## Warning: Removed 20799 rows containing non-finite outside the scale range
## (‘stat_boxplot()‘).
```



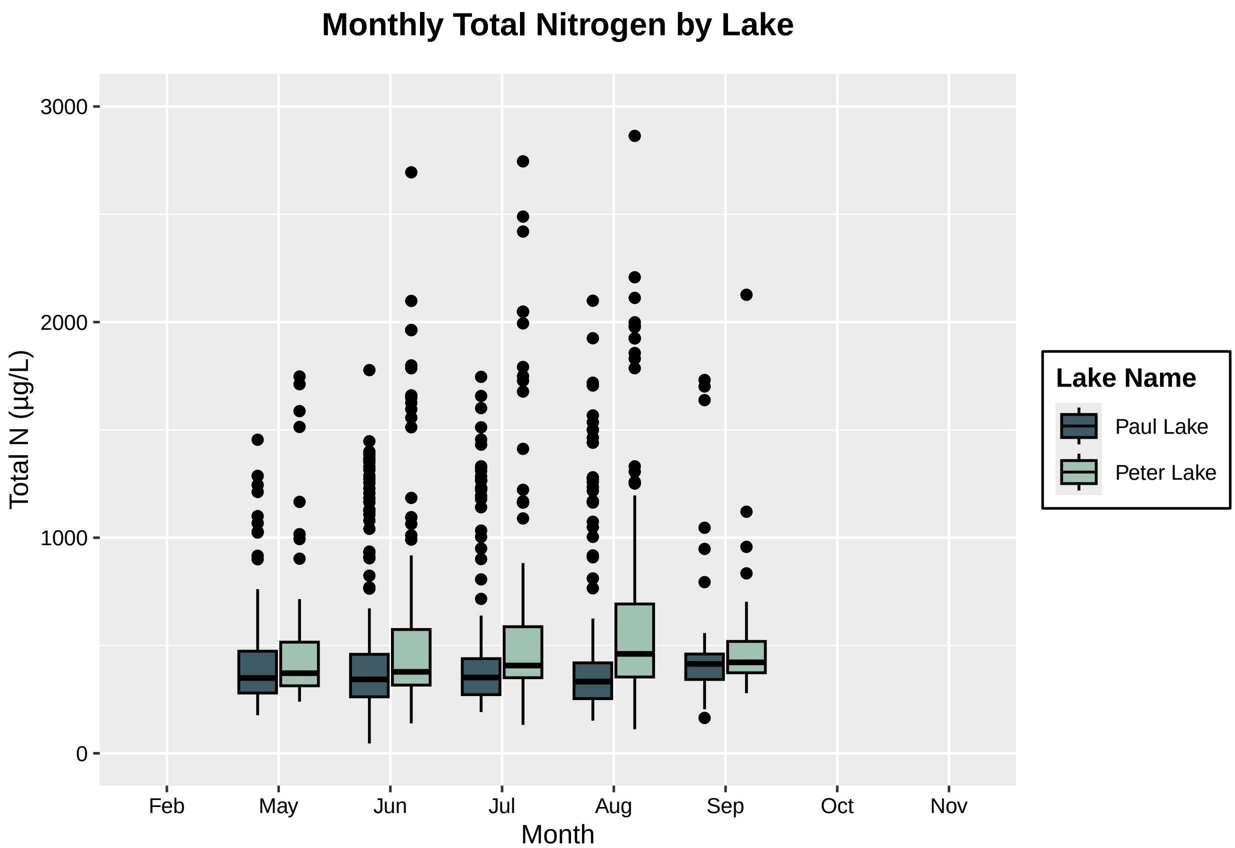

```r
summary(NTLLTER$tn_ug)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
##   45.67  300.00  380.58  508.55  523.48 3497.70   21583
```

```r
TN <- ggplot(NTLLTER, aes(x = month, y = tn_ug, fill = lakename)) +
  geom_boxplot(color = "black") +
  labs(
    title = 'Monthly Total Nitrogen by Lake',
    x = 'Month',
    y = 'Total N (µg/L)',
    fill = 'Lake Name'
  ) +
```

```
  scale_fill_manual(
    values = c("Peter Lake" = "#9FC2B2", "Paul Lake" = "#3F5B66")
  ) +
  ylim(0,3000) +
  Elizabeth_theme
print(TN)
```

```
## Warning: Removed 21585 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```




```
#start combining boxplots by extracting one legend from Temp
shared_legend <- get_legend(
  Temp +
    guides(fill = guide_legend(ncol = 1, title.position = "top")) +
    theme(legend.position = "right"))
```

```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning in get_plot_component(plot, "guide-box"): Multiple components found;
## returning the first one. To return all, use `return_all = TRUE`.
```

```r
#keep bottom x-axis title only
temp <- Temp + theme(legend.position = "none", axis.title.x = element_blank())
tp   <- TP   + theme(legend.position = "none", axis.title.x = element_blank())
tn   <- TN   + theme(legend.position = "none")

#Combine three plots and align axes
plots_col <- plot_grid(
  temp, tp, tn,
  ncol = 1,
  align = "v",
  axis = "l",
  rel_heights = c(2, 2, 2))
```

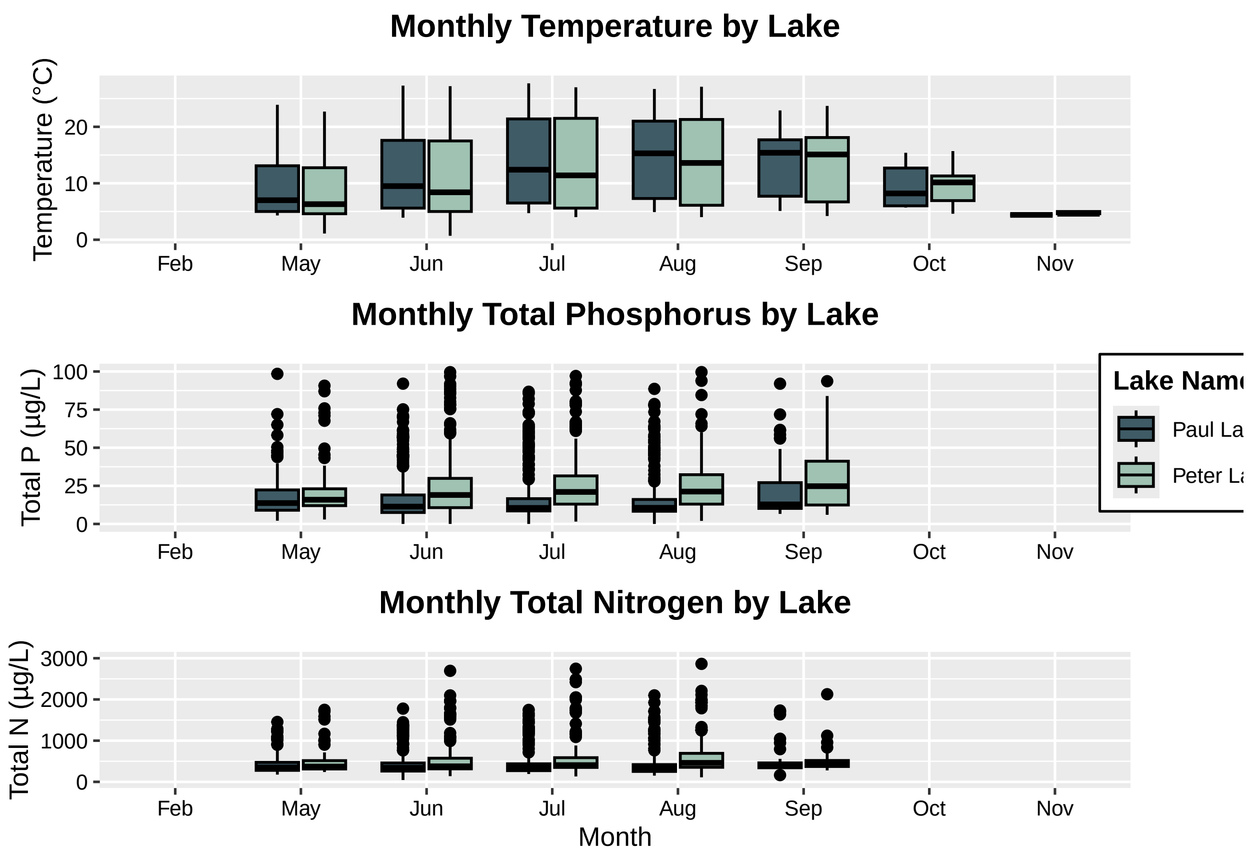
```
## Warning: Removed 3566 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 20799 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```
## Warning: Removed 21585 rows containing non-finite outside the scale range
## (`stat_boxplot()`).
```

```r
#attach the shared legend to the right of the stacked plots
CombinedPlots <- plot_grid(
  plots_col,
  ggdraw(shared_legend),
  ncol = 2,
  rel_widths = c(0.92, 0.08))

print(CombinedPlots)
```



Question: What do you observe about the variables of interest over seasons and between lakes?

> Answer: For both lakes, the highest temperatures are during the summer months (July and August) with a gradient decrease towards the winter months (May or Nov). The monthly total phosphorus and nitrogen are at higher concentrations and wider distribution (longer boxplot) in Peter lake than Paul lake. In particular, both lakes showcase a slight increase in monthly total phosphorus and nitrogen from May to September (with the exception of a sharper decrease in total nitrogen in September).

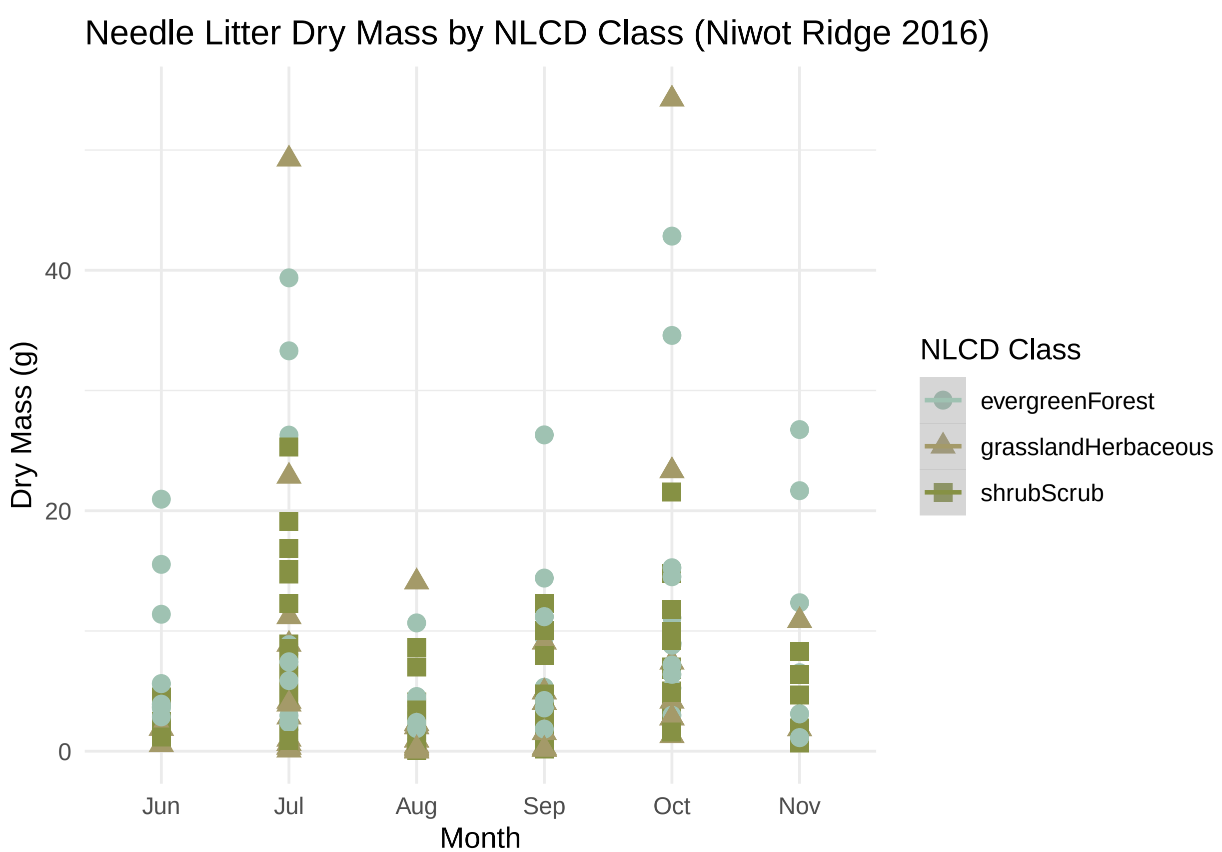
6. [Niwot Ridge] Plot a subset of the litter dataset by displaying only the "Needles" functional group. Plot the dry mass of needle litter by date and separate by NLCD class with a color aesthetic. (no need to adjust the name of each land use)
7. [Niwot Ridge] Now, plot the same plot but with NLCD classes separated into three facets rather than separated by color.

```
#6
NEON_NIWO <- NEON_NIWO %>%
  mutate(
    month = factor(
      month(collectDate),
      levels = 1:12,
      labels = month.abb,
      ordered = TRUE))

Needle_Groups <- NEON_NIWO %>%
```

```r
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x = month, y = dryMass, color = nlcdClass, shape = nlcdClass)) +
  geom_point(size = 3) +
  geom_smooth(method = "lm", se = TRUE, linewidth = 0.8) +
  scale_color_manual(
    values = c(
      "shrubScrub" = "#869144",
      "evergreenForest" = "#9FC2B2",
      "grasslandHerbaceous" = "#A49A69")) +
  labs(
    title = "Needle Litter Dry Mass by NLCD Class (Niwot Ridge 2016)",
    x = "Month",
    y = "Dry Mass (g)",
    color = "NLCD Class",
    shape = "NLCD Class") +
  theme_minimal() +
  theme(legend.position = "right")
print(Needle_Groups)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

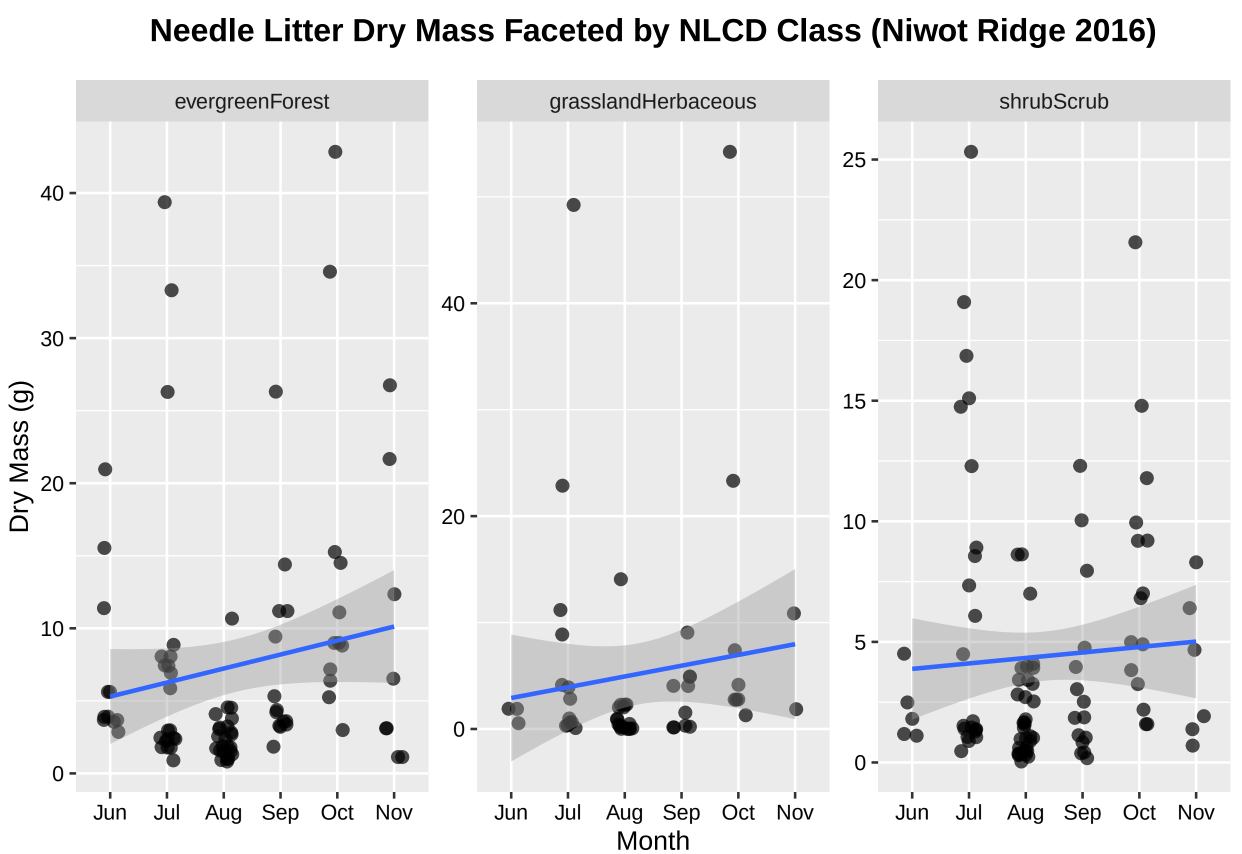



```r
#7

Needle_facets <- NEON_NIWO %>%
```

```r
  filter(functionalGroup == "Needles") %>%
  ggplot(aes(x = month, y = dryMass)) +
  geom_jitter(width = 0.15, height = 0, alpha = 0.7, size = 1.8) +
  geom_smooth(aes(group = 1), method = "lm", se = TRUE, linewidth = 0.8) +
  facet_wrap(~ nlcdClass, ncol = 3, scales = "free_y") +
  labs(
    title = "Needle Litter Dry Mass Faceted by NLCD Class (Niwot Ridge 2016)",
    x = "Month",
    y = "Dry Mass (g)"
  ) +
  Elizabeth_theme
print(Needle_facets)
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



Question: Which of these plots (6 vs. 7) do you think is more effective, and why?

> Answer: Plot 7 is more effective because you are able to see the dry mass values for each needle class type when NLCD classes are seperated into three facets. Plot 6 is harder to see, not because of the colors nor the point sizes, because each class overlaps one another on the plot. With Plot 7, we can create a trendline and make more confident conclusions about the relationship between time and dry mass for each NLCD class.